

security test bed to life. Navigate to the directory where you saved the file and open a terminal window. Docker Compose simplifies the process of running multiple containers by allowing you to use a single command to start everything up.

To start the containers for Kali Linux, DVWA, and Metasploitable, use the following command in the Windows command prompt from the directory where the *compose.yml* file is saved:

```
docker-compose up -d
```

This command will pull the necessary images from Docker Hub (if not already on your system), create the containers, and start them in detached mode (i.e., in the background) as illustrated in Figure 1. Each container will be set up according to the configuration in the *compose.yml* file, with the appropriate network settings, port mappings, and volumes.

The actions done in this step are:

- Docker pulls the specified images for Kali Linux, DVWA, Snort3 and Metasploitable.
- Containers are created and started based on the configurations in the *compose.yml* file.
- Each container is networked with the other one so they can interact with each other as needed.
- Services (like the Kali GUI or DVWA web application) are made accessible via your browser or terminal.

Docker Compose makes it easy to manage these containers. You can bring down the entire setup with a single command (*docker-compose down*), restart services, or scale the environment by adding more containers.

```
version: '3'
```

```
services:
```

```
  dvwa:
```

```
    image: vulnerables/web-dvwa
```

```
    container_name: dvwa
```

```
  ports:
```

```
    - "8080:80"
```

```
  networks:
```

```
    - pentest-network
```

```
  tty: true
```

```
  stdin_open: true
```

```
  restart: unless-stopped
```

```
  depends_on:
```

```
    - dvwa_db
```

```
  dvwa_db:
```

```
    image: mariadb:10.1
```

```
    container_name: dvwa_db
```

```
    hostname: dvwa_db
```

```
    volumes:
```

```
      - dvwa_db_data:/var/lib/mysql
```

```
  environment:
```

```
    MYSQL_ROOT_PASSWORD: rootpass
```

```
    MYSQL_DATABASE: dvwa
```

```
    MYSQL_USER: dvwa
```

```
    MYSQL_PASSWORD: p@ssw0rd
```

```
  restart: unless-stopped
```

```
  networks:
```

```
    - pentest-network
```

```
  kali:
```

```
    build:
```

```
      context: .
```

```
      dockerfile: Dockerfile.kali
```

```
    container_name: kali
```

```
    shm_size: '512m'
```

```
    ports:
```

```
      - "7002:6901"
```

```
    environment:
```

```
      - VNC_PW=password
```

```
    user: "root"
```

```
    networks:
```

```
      - pentest-network
```

```
    tty: true
```

```
    stdin_open: true
```

```
    restart: unless-stopped
```

```
    volumes:
```

```
      - kali_data:/root
```

```
    command: >
```

```
      bash -c "apt-get update && apt-
get install -y iputils-ping traceroute
whois net-tools nmap tshark wireshark
hping3 nano && tail -f /dev/null"
```

```
  metasploitable:
```

```
    build:
```

```
      context: .
```

```
    dockerfile: Dockerfile.
```

```
  metasploitable
```

```
    container_name: metasploitable
```

```
    networks:
```

```
      - pentest-network
```

```
    tty: true
```

```
    stdin_open: true
```

```
    restart: unless-stopped
```

```
    ports:
```

```
      - "7003:80"
```

```
    volumes:
```

```
      - metasploitable_data:/var/lib/
```

```
  metasploitable
```

```
  snort3:
```

```
    build:
```

```
      context: .
```

```
      dockerfile: Dockerfile.snort3
```

```
    container_name: snort3
```

```
    hostname: snort3
```

```
    user: "snorty"
```

```
    working_dir: /home/snorty
```

```
    networks:
```

```
      - pentest-network
```

```
    tty: true
```

```
    stdin_open: true
```

```
    restart: unless-stopped
```

```
    volumes:
```

```
      - snort_data:/etc/snort
```

```
    command: >
```

```
      bash -c "apt-get update && apt-
get install -y iputils-ping traceroute
whois net-tools nmap tshark wireshark
hping3 nano && tail -f /dev/null"
```

```
  networks:
```

```
    pentest-network:
```

```
      driver: bridge
```

```
  volumes:
```

```
    dvwa_db_data:
```

```
    kali_data:
```

```
    metasploitable_data:
```

```
    snort_data:
```

This repository sets up a self-contained penetration testing environment using Docker Compose. The lab includes four major components -- DVWA, Kali Linux, Metasploitable, and Snort3—designed to simulate a real-world attack-defence ecosystem.